

Introduction

In this lab, we explore the integration of RabbitMQ with gRPC to simulate a distributed system for mine detection and disposal. Building on the previous labs, the ground control still manages the environment, and the rovers continue to explore the land. However, unlike Lab 1 and Lab 2, the rovers no longer attempt to dig up mines. Instead, they act as reconnaissance units, detecting mines and publishing demining tasks to a RabbitMQ message queue. A new entity, the deminers, is introduced, which subscribes to these tasks and handles the mine disarming process asynchronously.

The primary goal of this lab is to implement a publisher-subscriber model using RabbitMQ, demonstrating how distributed systems handle task delegation and workload balancing. This lab emphasizes message-driven architecture, ensuring efficient communication between multiple entities while maintaining system scalability and reliability.

Results

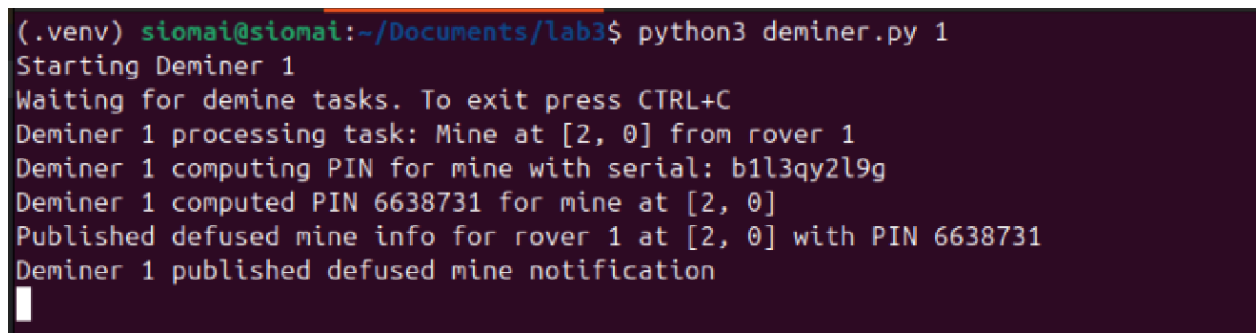
Here is the output of the three files (Ground Control Server, Deminers 1 & 2, and Rover Client), when Rover 1 is running. Additionally, there are two text files created, the path text file for each rover, and a defused mines log file where it contains all the defused mines.

Ground Control Server



```
Run gc_server x
/home/siomai/Documents/lab3/.venv/bin/python /home/siomai/Documents/lab3/gc_server.py
/home/siomai/Documents/lab3/.venv/lib/python3.12/site-packages/google/protobuf/runtime_version.py:98: UserWarning: Protobuf gencode version 3.21.12 is deprecated, please upgrade to 3.21.12 or later.
  warnings.warn(
Ground Control gRPC server started on port 50051.
Listening for defused mine notifications...
Waiting for defused mine notifications. To exit press CTRL+C
Defused mine notification received: {'coordinates': [2, 0], 'rover_id': 1, 'serial': 'b1l3qy2l9g', 'pin': 6638731}
Defused mine notification received: {'coordinates': [0, 1], 'rover_id': 1, 'serial': 'b1l3qy2l9g', 'pin': 6638731}
Rover 1 reported status: SUCCESS
```

Deminer 1



```
(.venv) siomai@siomai:~/Documents/lab3$ python3 deminer.py 1
Starting Deminer 1
Waiting for demine tasks. To exit press CTRL+C
Deminer 1 processing task: Mine at [2, 0] from rover 1
Deminer 1 computing PIN for mine with serial: b1l3qy2l9g
Deminer 1 computed PIN 6638731 for mine at [2, 0]
Published defused mine info for rover 1 at [2, 0] with PIN 6638731
Deminer 1 published defused mine notification
```

Deminer 2

```
(.venv) siomai@siomai:~/Documents/lab3$ python3 deminer.py 2
Starting Deminer 2
Waiting for demine tasks. To exit press CTRL+C
Deminer 2 processing task: Mine at [0, 1] from rover 1
Deminer 2 computing PIN for mine with serial: b1l3qy2l9g
Deminer 2 computed PIN 6638731 for mine at [0, 1]
Published defused mine info for rover 1 at [0, 1] with PIN 6638731
Deminer 2 published defused mine notification
```

Rover Client Path 1

```
(.venv) siomai@siomai:~/Documents/lab3$ python3 rover_client.py 1
/home/siomai/Documents/lab3/.venv/lib/python3.12/site-packages/google/protobuf/runtime_version.py:98: UserWarning: Protobuf gencode version 5.29.0 is exactly one major version older than the runtime version 6.30.0 at rover.proto. Please update the gencode to avoid compatibility violations in the next runtime release.
  warnings.warn(
Rover 1 received map of size 4x3
Rover 1 executing command: M
Rover 1 moved to position: (1, 0)
Rover 1 executing command: M
Rover 1 moved to position: (2, 0)
Rover 1 found a mine at (2, 0)!
Rover 1 received mine serial: b1l3qy2l9g
Published demine task for rover 1 at (2, 0)
Rover 1 published demine task for mine at (2, 0)
Rover 1 executing command: M
Rover 1 moved to position: (3, 0)
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: R
Rover 1 turned right, now facing W
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: L
Rover 1 turned left, now facing S
Rover 1 executing command: R
```

```
Rover 1 turned right, now facing W
Rover 1 executing command: R
Rover 1 turned right, now facing N
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: M
Rover 1 moved to position: (3, 1)
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: M
Rover 1 moved to position: (2, 1)
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: M
Rover 1 moved to position: (2, 2)
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: M
Rover 1 moved to position: (1, 2)
Rover 1 executing command: M
```

```
Rover 1 moved to position: (0, 2)
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: L
Rover 1 turned left, now facing W
Rover 1 executing command: M
Rover 1 moved to position: (0, 1)
Rover 1 found a mine at (0, 1)!
Rover 1 received mine serial: b1l3qy2l9g
Published demine task for rover 1 at (0, 1)
Rover 1 published demine task for mine at (0, 1)
Rover 1 executing command: M
Rover 1 moved to position: (0, 0)
Rover 1 executing command: R
Rover 1 turned right, now facing N
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: L
Rover 1 turned left, now facing W
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: L
```

```
Rover 1 turned left, now facing S
Rover 1 executing command: M
Rover 1 moved to position: (1, 0)
Rover 1 executing command: M
Rover 1 moved to position: (2, 0)
Rover 1 executing command: M
Rover 1 moved to position: (3, 0)
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: L
Rover 1 turned left, now facing E
Rover 1 executing command: R
Rover 1 turned right, now facing S
Rover 1 executing command: L
Rover 1 turned left, now facing E
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: M
Rover 1 moved to position: (3, 1)
Rover 1 executing command: M
```

```
Rover 1 moved to position: (3, 2)
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: M
Rover 1 moved to position: (2, 2)
Rover 1 executing command: L
Rover 1 turned left, now facing W
Rover 1 executing command: R
Rover 1 turned right, now facing N
Rover 1 executing command: M
Rover 1 moved to position: (1, 2)
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: M
Rover 1 moved to position: (0, 2)
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: M
Movement out of bounds! Staying at the same position.
Rover 1 executing command: R
```

```
Rover 1 turned right, now facing S
Rover 1 executing command: M
Rover 1 moved to position: (1, 2)
Rover 1 executing command: R
Rover 1 turned right, now facing W
Rover 1 executing command: R
Rover 1 turned right, now facing N
Rover 1 executing command: M
Rover 1 moved to position: (0, 2)
Rover 1 executing command: L
Rover 1 turned left, now facing W
Rover 1 executing command: R
Rover 1 turned right, now facing N
Rover 1 executing command: L
Rover 1 turned left, now facing W
Rover 1 executing command: L
Rover 1 turned left, now facing S
Rover 1 executing command: R
Rover 1 turned right, now facing W
Rover 1 executing command: M
Rover 1 moved to position: (0, 1)
Rover 1 executing command: R
Rover 1 turned right, now facing N
Rover 1 executing command: M
```

```
Movement out of bounds! Staying at the same position.
```

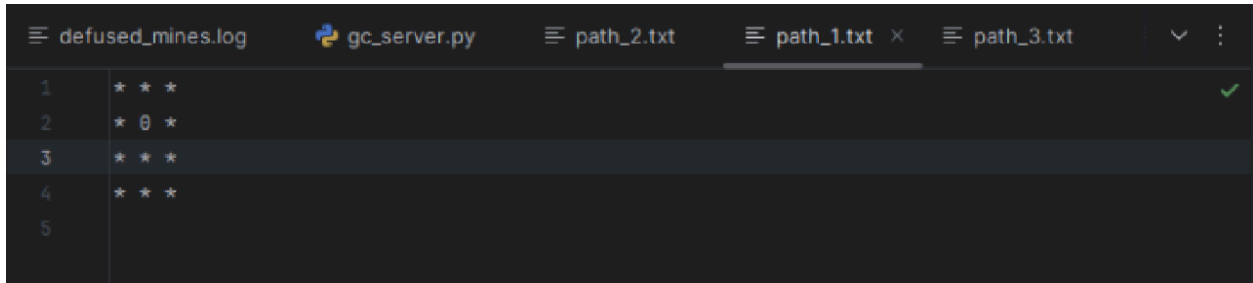
```
Rover 1 executing command: R
Rover 1 turned right, now facing E
Rover 1 executing command: L
Rover 1 turned left, now facing N
Rover 1 executing command: M
```

```
Movement out of bounds! Staying at the same position.
```

```
Rover 1 executing command: L
Rover 1 turned left, now facing W
Rover 1 executing command: M
Rover 1 moved to position: (0, 0)
Rover 1 path written to path_1.txt
Rover 1 finished execution.
```

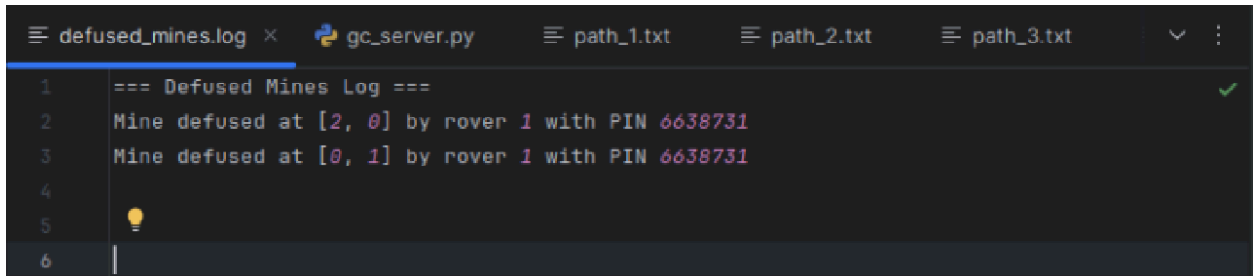
```
(.venv) siomai@siomai:~/Documents/lab3$
```

Path 1 text file:



```
defused_mines.log  gc_server.py  path_2.txt  path_1.txt x  path_3.txt  v  ...
1  * * *
2  * 0 *
3  * * *
4  * * *
5
```

Defused Mines log file:



```
defused_mines.log x  gc_server.py  path_1.txt  path_2.txt  path_3.txt  v  ...
1  === Defused Mines Log ===
2  Mine defused at [2, 0] by rover 1 with PIN 6638731
3  Mine defused at [0, 1] by rover 1 with PIN 6638731
4
5  💡
6
```

Conclusion

This lab successfully demonstrated how RabbitMQ can be used to facilitate distributed task processing in a multi-agent system. By separating exploration and demining tasks, we improved modularity and reduced the computational burden on individual components. The implementation of message queues allowed seamless task distribution between rovers and deminers, ensuring fault tolerance and asynchronous execution.

Through this exercise, we gained practical experience with RabbitMQ's pub-sub model, gRPC communication, and the challenges of designing scalable distributed systems. In a real-world scenario, such a system could be extended for autonomous robotic exploration, disaster recovery, or industrial automation. Future improvements could include load balancing for deminers, prioritization of high-risk mines, and real-time monitoring dashboards for better system observability.